Chandra Dhakal, PhD

Health Economist | Data to Policy Lead, CDC Foundation

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Professional Summary

PhD health economist with 10+ years of experience turning large, complex data into evidence that drives health and economic policy. Lead the Data to Policy program at the CDC Foundation (contract to CDC), partnering with ministries of health across 19 countries under the Bloomberg Philanthropies Data for Health Initiative. Core expertise in economic evaluation (cost-effectiveness, cost-utility, and budget impact analysis), decision-analytic modeling (Markov models, microsimulation, decision trees), and causal inference with quasi-experimental methods on large-scale administrative and survey data. Five years of hands-on analysis of medical and pharmacy claims data alongside actuarial teams, identifying cost and trend drivers that inform affordability and medical-cost strategy. Author of 25+ peer-reviewed publications in *JAMA Network Open*, the *Journal of Policy Analysis and Management*, and the *American Journal of Preventive Medicine*, among others.

Education

Ph.D., Applied Economics Aug 2022
University of Georgia, Athens, GA
Dissertation: "Unemployment Insurance, Food Preference Changes, and Health Outcomes." Advisor: Chen Zhen.

M.S., Statistics Aug 2021
University of Georgia, Athens, GA

M.S., Applied Economics Aug 2016
Texas Tech University, Lubbock, TX

Professional Experience

Health Economist, Data to Policy Lead Dec 2021 – Present
CDC Foundation (contract to CDC), Division of Global Health Protection, Center for Global Health, Atlanta, GA

- Lead the Data to Policy program under the Bloomberg Philanthropies Data for Health Initiative, working with ministries of health across 19 countries to strengthen evidence-based priority setting and resource allocation.
- Design and lead economic evaluations — cost-effectiveness, cost-utility, and budget impact analyses — and decision-analytic models (Markov models, microsimulation, decision trees) assessing CDC investments in global health protection and security.
- Develop and apply econometric, quasi-experimental, and machine learning methods to large administrative, household, and retail scanner datasets to inform food, health, and related economic outcomes.
- Build global capacity: design and deliver international training-of-trainers workshops in health economics and decision modeling for 100+ health economists and policymakers.

Research Associate Aug 2016 – Dec 2021
Department of Agricultural and Applied Economics, University of Georgia, Athens, GA

- Estimated structural demand systems (Exact Affine Stone Index) on retail scanner data with billions of observations.
- Compiled and analyzed IRI InfoScan data to create small-area price indexes for all U.S. counties within the Administrative Data Research Facilities, cutting processing time by 50%.
- Predicted type 2 diabetes using random forest and classification trees (75–79% accuracy); identified glucose, BMI, age, and pedigree as leading risk factors.
- Collaborated with the Centers for Medicare & Medicaid Services, contributing economic and evaluation expertise to federal health policy work.

Economist Jan 2015 – Jul 2016
Texas Tech University, Lubbock, TX

- Applied econometric and dynamic optimization models to agricultural programs; determined the economically optimal nitrogen rate for cotton (40 lb per bale, net present value $6,519).

Assistant Professor of Agricultural Economics Jan 2014 – Dec 2014
Tribhuvan University, Kathmandu, Nepal

Research Associate / Data Analyst Jan 2012 – Dec 2013
Nepal Development Research Institute, Kathmandu, Nepal

- Compiled, analyzed, and reported on public administrative datasets on health-related issues to explain and forecast economic trends for senior management.

Peer-Reviewed Publications

Neupane, S., W. J. Florkowski, and C. Dhakal. (2024). Trends and disparities in diabetes prevalence in the United States from 2012 to 2022. *American Journal of Preventive Medicine*, 67(2), 299–302.

Neupane, S., W. J. Florkowski, and C. Dhakal. (2025). Heterogeneous association between e-cigarette use and diabetes prevalence among US adults. *AJPM Focus*, 4(3), 100281.

Neupane, S., W. J. Florkowski, U. Dhakal, and C. Dhakal. (2024). Regional disparities in type 2 diabetes prevalence and associated risk factors in the United States. *Diabetes, Obesity & Metabolism*, 26(10).

Rezayi, S., Z. Liu, Z. Wu, C. Dhakal, B. Ge, H. Dai, G. Mai, N. Liu, C. Zhen, T. Liu, and S. Li. (2024). Exploring new frontiers in agricultural NLP: Investigating the potential of large language models for food applications. *IEEE Transactions on Big Data*.

Singh, A. K., B. Lamichhane, S. Devkota, U. Dhakal, and C. Dhakal. (2024). Do large language models show human-like biases? Exploring confidence–competence gap in AI. *Information*, 15(2), 92.

Dhakal, C., Y. Luo, S. Wang, and C. Zhen. (2024). Unemployment insurance benefit reduction and food hardship. *Journal of Policy Analysis and Management*, 43(2), 530–554.

Acharya, B., D. Bhatta, and C. Dhakal. (2022). The risk of eviction and the mental health outcomes among US adults. *Preventive Medicine Reports*, 29, 101981.

Acharya, B. and C. Dhakal. (2022). Risky health behaviors during the COVID-19 pandemic: Evidence from the expenditures on alcohol, non-alcoholic beverages, and tobacco products. *PLOS ONE*. (Co-first authorship)

Dhakal, C., B. Acharya, and S. Wang. (2022). Food spending in the United States during the first year of the COVID-19 pandemic. *Frontiers in Public Health*, 10.

Dhakal, C. and C. L. Escalante. (2022). The productivity effects of adopting improved organic manure practices in Nepal. *Frontiers in Environmental Science*.

Rezayi, S., Z. Liu, Z. Wu, C. Dhakal, B. Ge, C. Zhen, T. Liu, and S. Li. (2022). AgriBERT: Knowledge-infused agricultural language models for matching food and nutrition. *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)*.

Dhakal, C., S. Khadka, C. Park, and C. L. Escalante. (2022). Climate change adaptation and its impact on household farm income and revenue risk exposure. *Resources, Environment and Sustainability*.

Acharya, B. and C. Dhakal. (2021). Implementation of state vaccine incentive lottery programs and uptake of COVID-19 vaccinations in the US. *JAMA Network Open*. (Co-first authorship)

Dhakal, C. and S. Khadka. (2021). Heterogeneities in consumer diet quality and health outcomes of consumers by store choice and income. *Nutrients*, 13, 1046.

Joshi, R. and C. Dhakal. (2021). Predicting type 2 diabetes using logistic regression and machine learning approaches. *International Journal of Environmental Research and Public Health*, 18, 7346. (Co-first authorship)

Dhakal, C. and K. Lange. (2021). Crop yield response functions to nutrient application. *Agronomy Journal*.

Dhakal, C., C. L. Escalante, and C. Dodson. (2019). Heterogeneity of farm loan packaging term decisions: A finite mixture approach. *Applied Economics Letters*, 26(18), 1528–1532.

Dhakal, C., P. P. Regmi, I. P. Dhakal, B. Khanal, U. K. Bhatta, S. R. Barsila, et al. (2013). Perception, impact and adaptation to climate change: An analysis of livestock system in Nepal. *Journal of Animal Science Advances*, 3(9), 462–471.

Dhakal, C., K. Lange, M. N. Parajulee, and E. Segarra. (2019). Dynamic optimization of nitrogen in plateau cotton yield functions with nitrogen carryover considerations. *Journal of Agricultural and Applied Economics*, 51(3), 385–401.

Working Papers and Under Review

Dhakal, C. and S. Neupane. How did the US household respond to the Economic Impact Payments of 2020? Evidence from the Consumer Expenditure Survey. (Working paper)

Dhakal, C. and S. Neupane. Monthly cash, food spending, and food security: Evidence from the 2021 Child Tax Credit expansion and its expiration. (Working paper)

Dhakal, C. and S. Neupane. SNAP emergency allotments and food security: Evidence from the 2020 expansion and the 2023 benefit cliff. (Working paper)

Dhakal, C., S. Neupane, U. Dhakal, and W. J. Florkowski. Rising coronary heart disease prevalence in the United States: Trends, risk factors, and disparities, 2015–2022. (Revise and resubmit)

Dhakal, C., S. Khadka, and C. Zhen. Impact of unemployment on health status, mental health, and healthcare utilization: Evidence from firm closure during COVID-19. (Under review)

Dhakal, C., Y. Luo, S. Wang, and C. Zhen. The impact of unemployment insurance benefit reduction on food expenditures and food hardship: Evidence from the Federal Pandemic Unemployment Compensation expiration. (Under review)

Wang, S. and C. Dhakal. Measuring the welfare effect of the Low-Income Home Energy Assistance Program using the Exact Affine Stone Index demand system. (Under review)

Price and wealth effects on the quality of food purchases (with C. Zhen).

Purchase efficiency and the usage of SNAP benefits (with C. Zhen and Smith).

Impact of stimulus payments on consumer spending (with S. Wang).

Teaching Experience

Instructor of Record, University of Georgia Spring 2019
Analytical and Computational Tools for Applied Economics

Teaching Assistant, University of Georgia 2018 – 2019
Principles of Microeconomics (Spring 2018); Futures and Options Markets (Fall 2019)

Selected Conference Presentations

Dhakal, C. and C. Zhen. The Great Recession and nutritional quality of food purchases. Agricultural and Applied Economics Association Annual Meeting, Kansas City, MO, July 2020.

Dhakal, C. Healthfulness of household food purchases based on the food retail environment. 89th Annual Conference of the Southern Economic Association, Fort Lauderdale, FL, November 2019.

Dhakal, C. Do food purchases by SNAP recipients vary with food groups over the benefit month? Agricultural and Applied Economics Association Annual Meeting, Atlanta, GA, July 2019.

Dhakal, C. Do food purchases by SNAP recipients vary in nutrition quality over the benefit month? 88th Annual Conference of the Southern Economic Association, Washington, DC, November 2018.

Dhakal, C. and C. Escalante. Heterogeneity of farm loan packaging term decisions: A finite mixture model approach. Agricultural and Applied Economics Association Annual Meeting, Washington, DC, August 2018.

Honors and Awards

- **Grant Recipient**, U.S. Agency for International Development (USAID) — $35,000.
- **2020 Outstanding Article Award**, *Journal of Agricultural and Applied Economics* — for "Dynamic Optimization of Nitrogen in Plateau Cotton Yield Functions with Nitrogen Carryover Considerations."
- **2019 Graduate School Travel Award**, University of Georgia
- **2018 Graduate School Travel Award**, University of Georgia
- **2015 Travel Award**, Colorado State University, Livestock Innovation Lab
- **2015 Study Abroad Competitive Scholarship**, Texas Tech University, Office of International Affairs
- **2014 Early Career Research Scholar**, Feed the Future Innovation Lab, USAID
- **2011 International Graduate Student Fellowship**, Feed the Future Innovation Lab, USAID

Professional Service

Peer Review. Reviewer for *JAMA Network Open*, *Health Services Research*, *BMJ*, *American Journal of Preventive Medicine*, *PLOS ONE*, *Applied Economics*, *Frontiers in Medicine*, *Econometrics*, *Sustainability*, and *Nutrients*.

Memberships. American Society of Health Economists (2020–present); Agricultural & Applied Economics Association (2015–present); Southern Economics Association (2017–present); Southern Agricultural Economics Association (2015–present); Agricultural Economics Society, UK (2020–present).

Technical Skills

- **Programming & Data:** R, Python (pandas, NumPy, scikit-learn), Stata, SAS, SQL, AWS, LaTeX
- **Visualization & BI:** Tableau, Power BI, Excel
- **Decision Modeling:** TreeAge; Markov models, microsimulation, decision-tree analysis, QALY/DALY estimation
- **Methods:** Cost-effectiveness / cost-utility / budget impact analysis; causal inference (difference-in-differences, instrumental variables, regression discontinuity, synthetic control, propensity score matching); machine learning; Bayesian statistics; survival analysis; forecasting; time series; text mining; experimental and quasi-experimental design

References

Available upon request.